

CSE33 I : Microprocessor Interfacing and Embedded Systems

Lecture#18: Analog-to-Digital Converter (ADC)

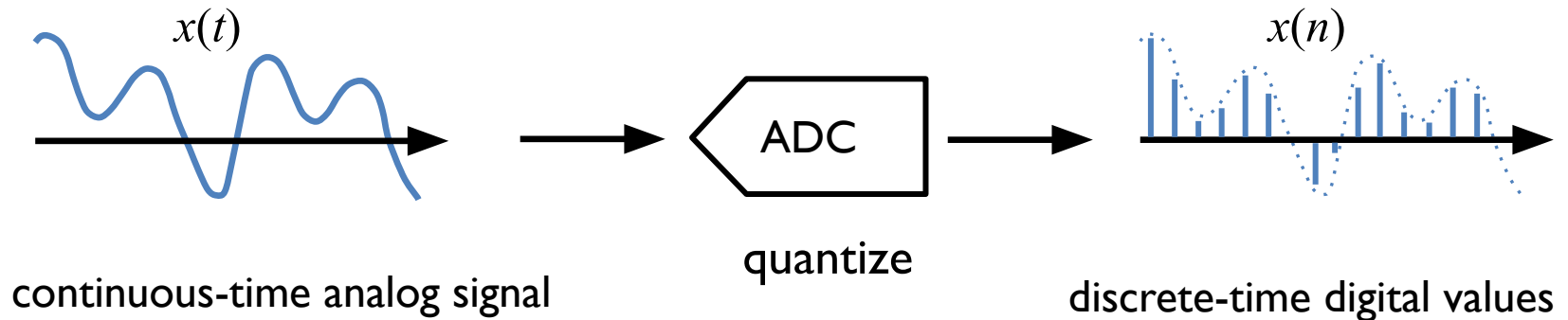
Chapter 20

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Analog-to-Digital Converter (ADC)

- ADC is important almost to all application fields
- Converts a continuous-time voltage signal within a given range to discrete-time digital values to quantify the voltage's amplitudes



Analog-to-Digital Converter (ADC)

- Three performance parameters:

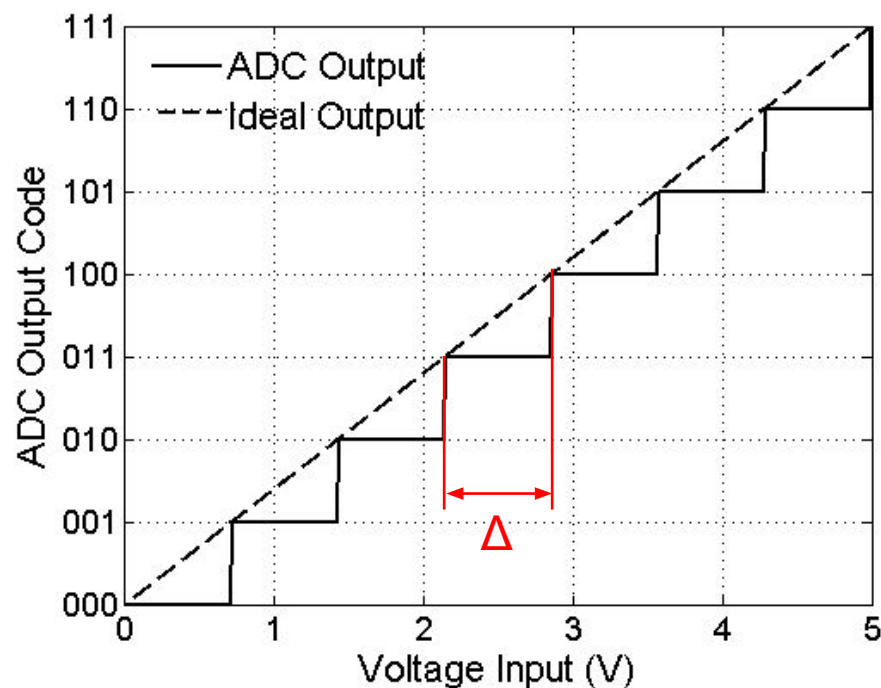
- sampling rate
- resolution
- power dissipation

- Many ADC implementations:

- sigma-delta
- successive-approximation
- pipeline

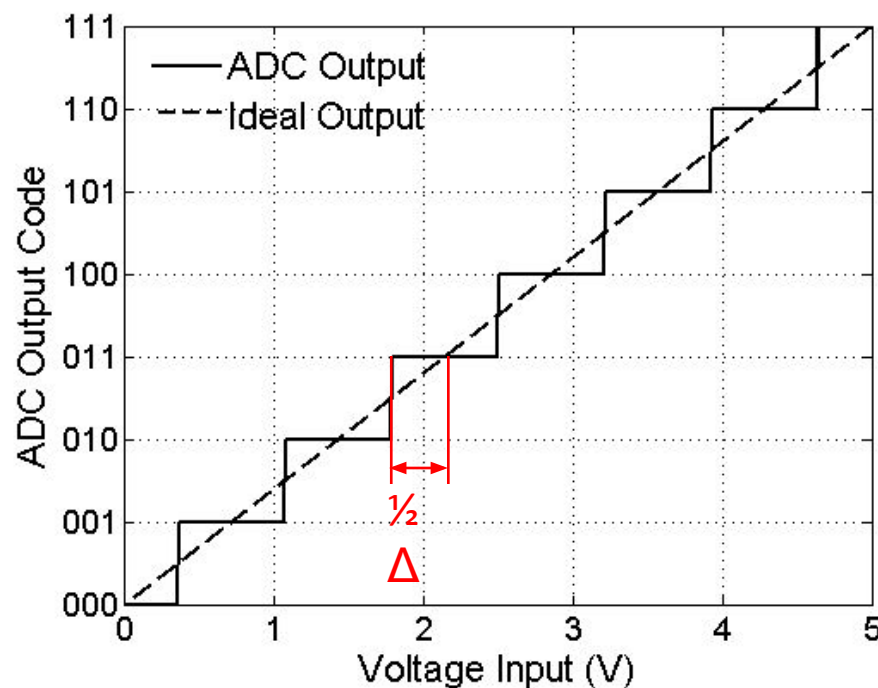
Resolution

- Resolution is determined by number of bits (in binary) to represent an analog input.
- Example of two quantization methods ($N = 3$)



$$\text{Digital Result} = \text{floor}\left(2^3 \times \frac{V}{V_{REF}}\right)$$

$$\text{Max quantization error} = \Delta = V_{REF}/2^3$$



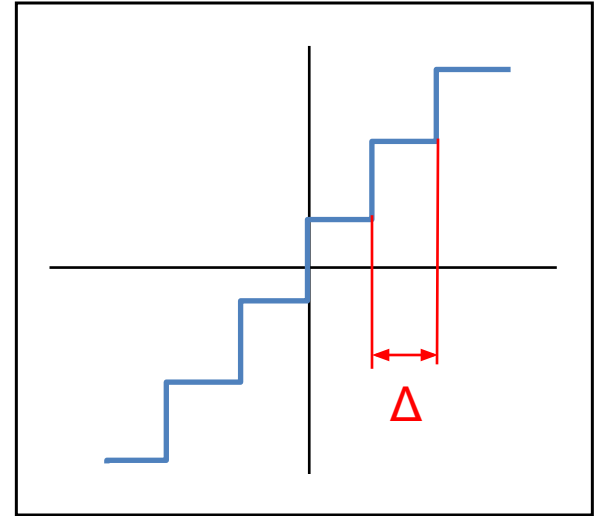
$$\text{Digital Result} = \text{round}\left(2^3 \times \frac{V}{V_{REF}}\right)$$

$$\text{Max quantization error} = \pm 1/2 \Delta = \pm V_{REF}/2^4$$

$$\text{round}(x) = \text{floor}(x + 0.5)$$

Quantization Error

- For N-bit ADC, it is limited to $\pm \frac{1}{2}\Delta$
- Δ = is the step size of the converter.

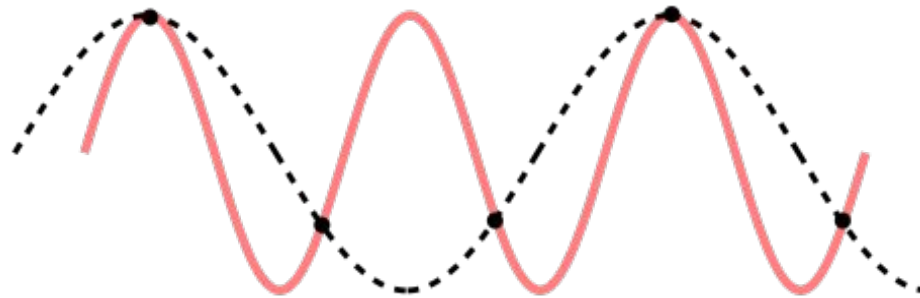


- Example: for 12-bit ADC and input voltage range [0, 3V]

$$\text{Max Quantization Error} = \frac{1}{2}\Delta = \frac{3V}{2 \times 2^{12}} = 0.367\text{mV}$$

Minimum Sampling Rate: Nyquist–Shannon Sampling Theorem

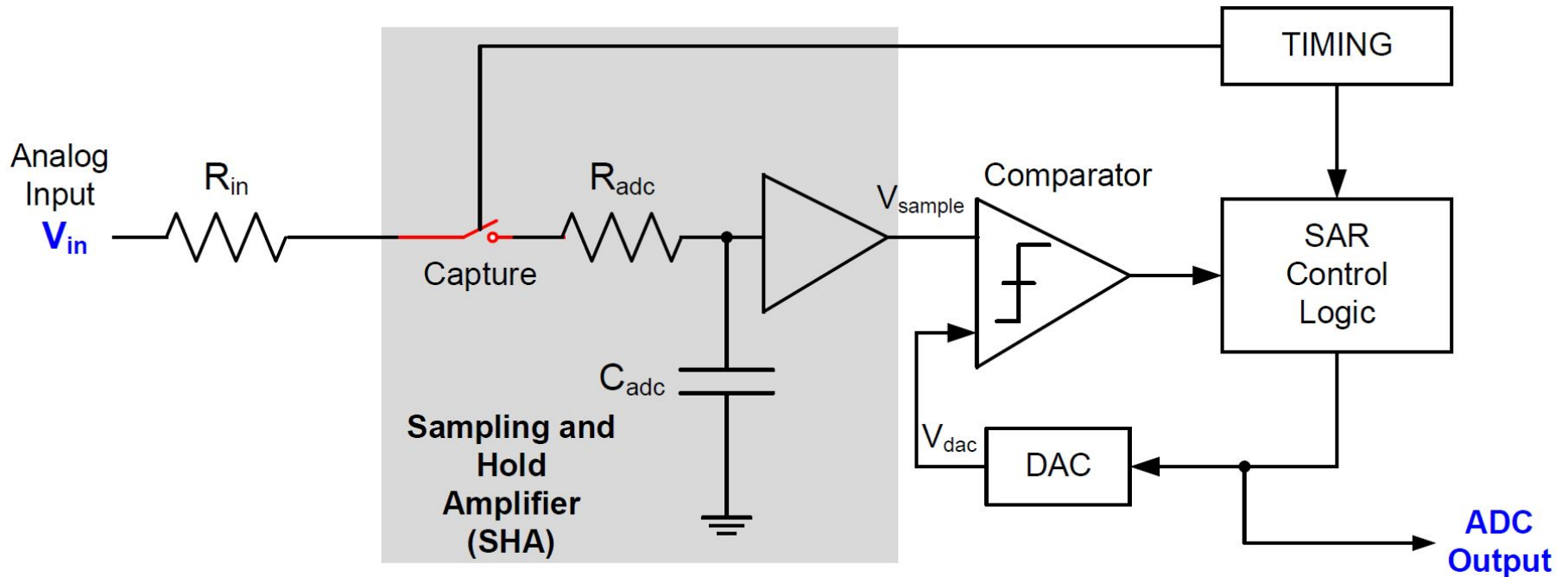
- In order to be able to reconstruct the analog input signal, the sampling rate should be at least twice the maximum frequency component contained in the input signal
- Example of two sine waves have the same sampling values. This is called **aliasing**.



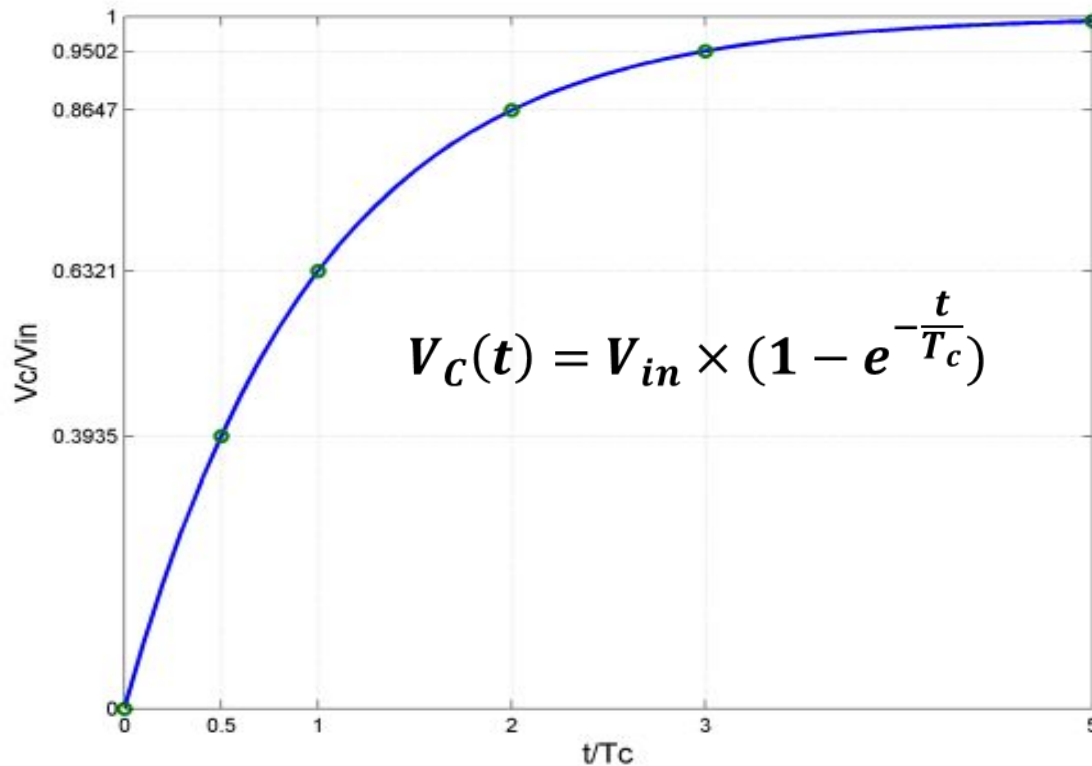
from
wiki.com

- **Antialiasing** (beyond the scope of this course)
 - **Pre-filtering**: use analog hardware to filtering out high-frequency components and only sampling the low-frequency components. The high-frequency components are ignored.
 - **Post-filtering**: Oversample continuous signal, then use software to filter out high-frequency components

Successive-approximation (SAR) ADC



Determining Minimum Sampling Time

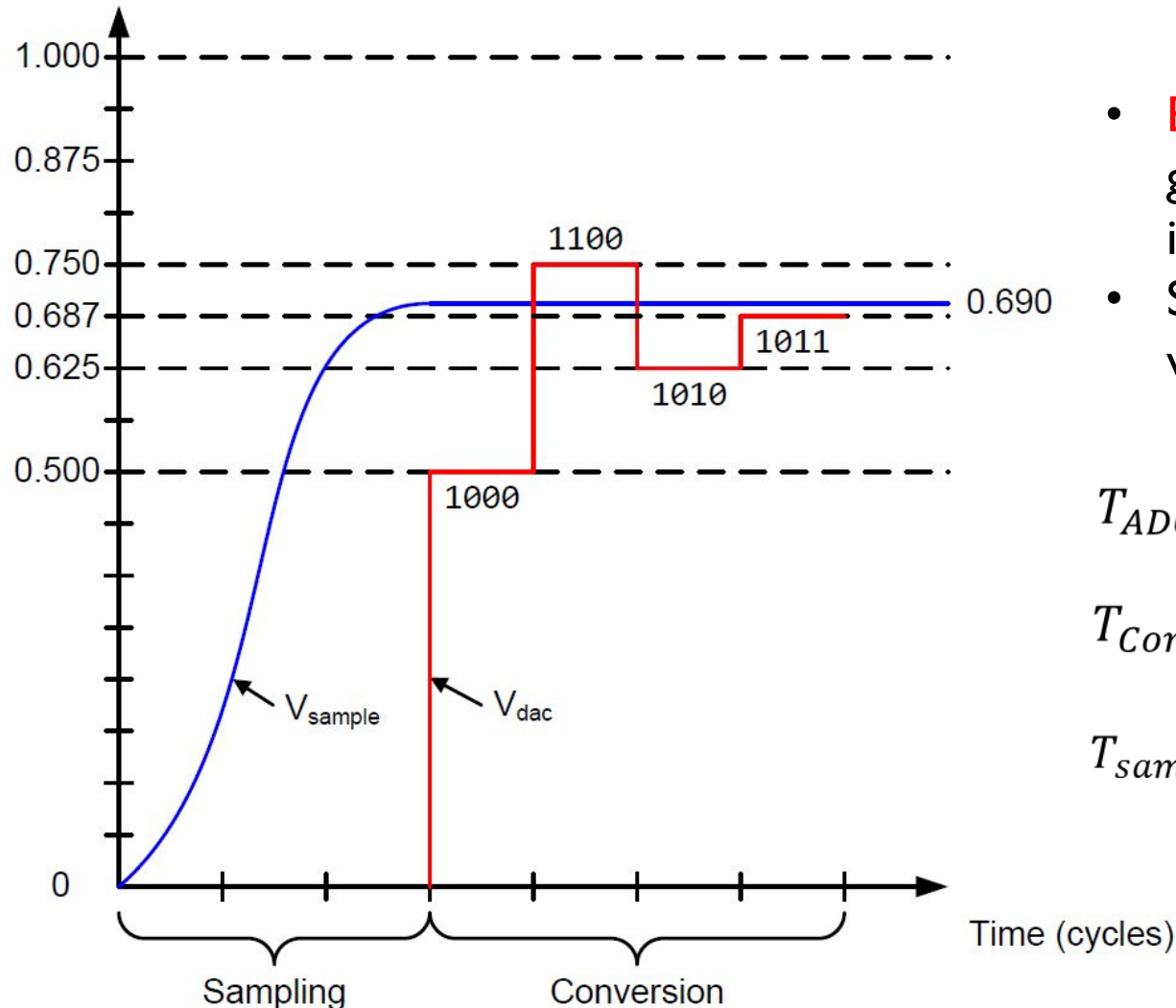


Sampling time is software programmable!

Larger sampling time $\begin{cases} \rightarrow \text{Smaller sampling error} \\ \rightarrow \text{Slower ADC speed} \end{cases}$

Tradeoff

Successive-approximation (SAR) ADC



- **Binary search** algorithm to gradually approaches the input voltage
- Settle into $\pm 1/2$ LSB bound within the time allowed

$$T_{ADC} = T_{\text{sampling}} + T_{\text{Conversion}}$$

$$T_{\text{Conversion}} = N \times T_{\text{ADC_Clock}}$$

T_{sampling} is software configurable

□ END OF CLASS

